PRACTICAL – 11

1. Aim

Interface Fingerprint Sensor and 16X2 LCD with Raspberry and display message Fingerprint matched or unmatched.

2. Prior Concepts

i) Concept of Python Programming.

ii) Knowledge of UART serial communication.

iii) Basics of I2C communication.

iv) Basic LCD display programming.

3. New Concepts

i) Interfacing a biometric fingerprint sensor with Raspberry Pi.

ii) Using UART communication for fingerprint sensor data exchange.

iii) Interfacing a 16×2 LCD through the I2C protocol.

iv) Developing a complete fingerprint-based access verification system.

4. Hardware Required

i) Fingerprint Sensor (R307, AS608, etc.)

ii) 16×2 LCD with I2C module

iii) Jumper Wires

5. Procedure

Pin Connection

i) Fingerprint Sensor → Raspberry Pi

Fingerprint Sensor	Raspberry Pi
VCC	5V
GND	GND
TX	GPIO15 (RXD)
RX	GPIO14 (TXD)

ii) I2C LCD → Raspberry Pi

LCD I2C	Raspberry Pi
VCC	5V
GND	GND
SDA	GPIO2 (SDA)
SCL	GPIO3 (SCL)

Input image

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Enable UART on Raspberry Pi:

sudo raspi-config

Navigate to:

Interface Options → Serial Port

Login Shell? → No

Serial Hardware? → Yes

Reboot:

sudo reboot

Enable I2C on Raspberry Pi :

sudo raspi-config

Navigate to:

Interface Options → I2C → Enable

Install Libraries

sudo apt update
sudo apt install python3-pip -y
pip3 install pyfingerprint
pip3 install RPLCD
pip3 install RPLCD-i2c

6. Implementation with Python Code

Code :

```
from pyfingerprint.pyfingerprint import PyFingerprint
from RPLCD.i2c import CharLCD
from time import sleep
```

```
# LCD configuration
lcd = CharLCD('PCF8574', 0x27)
lcd.clear()
lcd.write_string('Place Finger')

try:

    # Fingerprint sensor
    f = PyFingerprint('/dev/serial0', 57600, 0xFFFFFFFF,
                      0x00000000)

    if not f.verifyPassword():
        raise ValueError ('Fingerprint sensor password incorrect')

except Exception as e:

    print ('Sensor initialization failed!')

    print (e)

    exit (1)

while True:

    lcd.clear ()

    lcd.write_string ('Place Finger')

    # Wait for finger
    while not f.readImage():

        pass

    # Convert image
```

```
f.convertImage(0x01)

# Search template

result = f.searchTemplate()

positionNumber = result[0]

lcd.clear ()

if positionNumber >= 0:

    lcd.write_string('Fingerprint')

    lcd.crlf ()

    lcd.write_string('Matched')

    print ("Fingerprint Matched")

else:

    lcd.write_string('Fingerprint')

    lcd.crlf()

    lcd.write_string('Unmatched')

    print("Fingerprint Unmatched")

sleep(3)
```

7. Result